

INVESTORBENCH: A Benchmark for Financial Decision-Making Tasks with LLM-based Agent

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Abstract

Recent advancements have underscored the potential of large language model (LLM)-based agents in financial decision-making. Despite this progress, the field currently encounters two main challenges: (1) the lack of a comprehensive LLM agent framework adaptable to a variety of financial tasks, and (2) the absence of standardized benchmarks and consistent datasets for assessing agent performance. To tackle these issues, we introduce INVESTORBENCH, the first benchmark specifically designed for evaluating LLM-based agents in diverse financial decision-making contexts. INVESTORBENCH enhances the versatility of LLM-enabled agents by providing a comprehensive suite of tasks applicable to different financial products, including single equities like stocks, cryptocurrencies and exchange-traded funds (ETFs). Additionally, we assess the reasoning and decision-making capabilities of our agent framework using thirteen different LLMs as backbone models, across various market environments and tasks. Furthermore, we have curated a diverse collection of open-source, multi-modal datasets and developed a comprehensive suite of environments for financial decision-making. This establishes a highly accessible platform for evaluating financial agents' performance across various scenarios.

1 Introduction

The recent studies on large language model (LLM)-based agents have demonstrated impressive performance across a range of decision-making tasks in complex and open-ended environments spanning various domains (Zhang et al., 2024b; Guo et al., 2024; Eigner and Händler, 2024; Wang et al., 2024). However, developing agentic frameworks tailored specifically for financial decision-making remains a significant challenge. This complexity arises from the need for agents to acutely discern and prioritize decisive signals, and then make sequentially

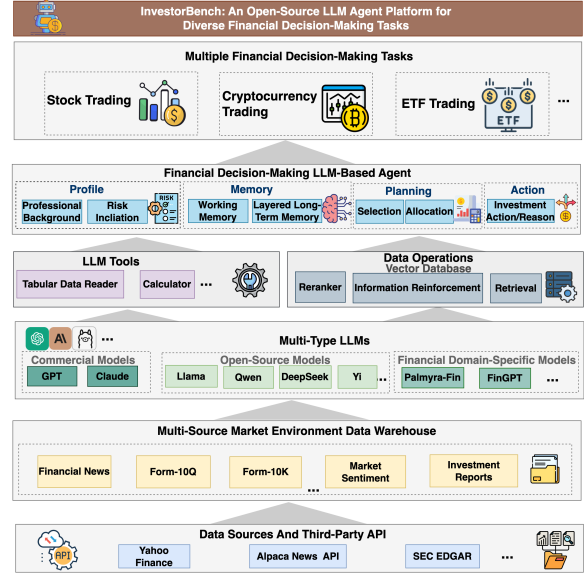


Figure 1: General architecture of INVESTORBENCH.

high-quality decisions within the volatile and multi-faceted financial markets, where information varies in time sensitivity and modality.

Furthermore, the design of financial agents becomes increasingly complex when applied across multiple decision-making tasks, due to the significant variation in key factors influencing financial decisions across different objectives and task types. For instance, single-equity tasks like stock trading require analyzing company-specific and industry-wide data, including market metrics, sector trends, performance reports, and relevant news (Yi et al., 2022). In contrast, cryptocurrency trading is highly sensitive to crypto-specific news and sentiment due to its dynamic nature (Bhatnagar et al., 2023). ETFs, on the other hand, typically follow passive investment strategies, emphasizing long-term growth and cost efficiency (Madhavan, 2016).

The recent emergence of financial LLM-based agent frameworks such as FINMEM (Yu et al., 2024a), FINAGENT (Zhang et al., 2024a), CRYPTO-

TRADE (Li et al., 2024), FINROBOT (Yang et al., 2024), and FINCON (Yu et al., 2024b) has presented a variety of architectural approaches tailored to specific financial tasks. This diversification has sparked substantial interest across both academic and industrial landscapes. FINROBOT is engineered specifically for market analysis, while FINMEM and FINAGENT are oriented towards trading individual equities like stocks and ETFs. CRYPTOTRADE focuses solely on cryptocurrency trading. FINCON pioneers in addressing portfolio management, although it currently handles only compact portfolios consisting of three stock assets. While these frameworks are effective within their respective niches, they generally focus on addressing only limited types of financial decision-making tasks. This restricts them from further demonstrating the broader applicability of these frameworks and limits the comprehensive, comparative insights that could be drawn from their overall decision-making performance. Furthermore, the frequent reliance on proprietary financial data complicates the evaluation of these tools, obscuring their effectiveness and adaptability in broader contexts. Therefore, there is a pressing need to develop innovative benchmarks specifically designed to evaluate LLM-based agents across a wider spectrum of financial decision-making scenarios. Such benchmarks would enable a more robust assessment of these technologies, facilitating advancements that could cater to various financial applications.

We introduce INVESTORBENCH, an open-source, LLM-based agent benchmark that generalizes across a broad range of financial decision-making tasks. Its detailed structure is illustrated in Figure 1. Further developed upon the foundational framework of FINMEM (Yu et al., 2024a), which focuses on single-stock investment decisions, our benchmark extends the scope to encompass an **ensemble of diverse financial market environments** for various financial tasks. INVESTORBENCH’s cognitive architecture, similar to FINMEM, employs a **layered memory processing** mechanism with distinct decay rates, enabling the agent to store, retrieve, and consolidate insights and reflections more effectively than the pure similarity-based memory retrieval used in FINAGENT. This approach ensures that decisions are informed by timely and impactful data, a capability previously shown effective for single-asset trading. These features reflect how human traders draw sequential decisions upon investment signals from multiple

sources and varying time sensitivities, allowing the agent to naturally adapt to complex financial tasks. INVESTORBENCH expands its evaluation beyond the original stock trading tasks to encompass three decision tasks significant in the realm of financial investment: **stock trading, cryptocurrency trading, and ETF investing**.

In summary, we make three key contributions: 1) We establish INVESTORBENCH, an **innovative and comprehensive financial agentic benchmark** designed to evaluate the reasoning and sequential decision-making capabilities of LLM-based agents in complex, open-ended financial scenarios. This benchmark provides a realistic perspective for assessing the design and performance of such agents. 2) We provide a **set of open-source, multi-source market environments** that closely mirror real-world conditions. Furthermore, these environments also serve as a standardized platform for evaluating the decision-making performance of other LLM-based financial agents. 3) We present a **unified, flexible language-agent framework** that allows finance professionals to conveniently customize assess any LLMs serving as the agent’s reasoning core. In this paper, we conduct a holistic evaluation of 13 LLMs — including recent, competitive, and domain-specific fine-tuned models (see Table 1) to provide a broad overview of their reasoning capabilities in sequential decision-making tasks within financial contexts.

2 LLM Trading Agents

In this section, we define a framework of the LLM-based agents in the INVESTORBENCH and formalize the financial decision-making tasks within the context of partially observable Markov decision process (POMDP) (Bertsekas and Shreve, 1996; Liu et al., 2020; Kabbani and Duman, 2022).

2.1 Definition

The LLM-based agent in INVESTORBENCH is structured as a large language model-modulo framework, designed to match or surpass the capabilities of professional human investors. This framework consists of several interconnected modules, each tailored to handle the distinct challenges presented by the financial market’s volatility and complexity: **Brain/Backbone (LLM)**: This module, which is the LLM itself, serves as the core of the LLM-based agent. It enhances the agent’s capabilities by enabling it to understand, process, and generate

natural language. This module plays a crucial role in supporting complex decision-making processes, offering interpretations of market-related information, generating predictive analytics, and reflecting on past investment decisions.

Perception: This module serves a critical function by converting raw market data into a structured format that is compatible with the LLM, specifying what the agent perceives and observes, which includes numerical, textual, and visual information.

Profile: This module serves two functions articulated in natural language. Firstly, it describes the agent’s role, highlighting its character as an experienced investor with expert-level knowledge and a self-adaptive risk preference. This risk preference dynamically adjusts based on historical market momentum, allowing the agent to optimize its strategies in real time. Secondly, the module provides a detailed background of the decision-making task, specifying the key characteristics and pertinent information about the target assets involved in the trading decisions, such as equity historical performance, price fluctuations, and sector information. This dual-function module supports the agent’s decisions with both the current market context and its historical performance.

Memory: This module processes and retains essential market data and historical insights, allowing the agent to draw on a rich repository of knowledge for decision-making. Building upon the pioneering work of Yu et al. (2024a) in FINMEM, the memory architecture comprises two primary components: **Working Memory** and **Layered Long-Term Memory**, as depicted in Figure 2.

Working memory: This component maintains FINMEM’s original functionalities: *observation*, *summarization*, and *reflection*. It incorporates two reflection mechanisms: *immediate* and *extended*. Immediate reflection produces the agent’s reasoning outcomes by integrating current market indicators with the top- K ranked events from each long-term memory layer, which are significant during both warm-up and evaluation stages. In the warm-up stage, the emphasis shifts as the trading direction is predetermined, focusing on understanding market trends and enhancing predictive accuracy. In the evaluation stage, it outputs the trading direction (Buy, Sell, or Hold), the rationale for this decision, identifying the most influential memory events and their respective IDs from each layer.

Layered Long-Term Memory: Inspired by the human cognitive system’s varying information decay speeds, Layered Long-Term Memory component structures financial insights across multiple layers. Each layer is represented by a vector database in the Long-Term Memory data warehouse, where information is prioritized and purged based on a specific decay rate. Deeper layers retain information longer with smaller decay rates, while shallower layers, dealing with more transient data, have larger decay rates. This tiered approach is critical as it allows the adaptation of the memory architecture to a broader range of financial tasks beyond single-asset decisions, accommodating an expanded variety of data sources and increasing overall system flexibility. Detailed mechanisms for ranking and decay in each layer are further elaborated in the Appendix A.

Action: This module executes trading and investment decisions based on the analysis provided by other modules. It directly outputs {“Buy”, “Sell”, “Hold”} for traded asset (stock, crypto, or ETF), as instructed by the backbone LLM. Action module synthesizes the operational outcomes from the Profile and Memory modules to facilitate precise and well-informed investment decisions. For its daily trading operations, the agent can choose from three specific actions for the traded asset: “Buy”, “Sell”, or “Hold”. The functionality and input requirements of this module differ significantly between the warm-up and evaluation stages: *during the warm-up stage*, the agent observes daily adjusted price differences between consecutive days, which are critical for identifying potential “Buy” or “Sell” signals. This period allows the agent to calibrate and adjust its decision-making strategies based on near-term market movements; *during the evaluation stage*, access to future price data is restricted, compelling the agent to rely solely on available historical data and its cognitive processing capabilities. In response to trading inquiries, the module integrates historical Profit & Loss (PnL), outcomes from extended reflections, and the top- K retrieved memories. This integration ensures that each trading decision is grounded in a comprehensive analysis of past performance and current market conditions.

2.2 Modeling financial decision-making

Formally, we model a financial decision-making process as infinite horizon POMDP with time index $\mathbb{T} = \{0, 1, 2, \dots\}$ and discount factor $\alpha \in (0, 1]$.